Beyond the Delay: Frequency, Impact and Preventibility in 2025

The team

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Introduction

Toronto's subway system is an integral component of daily life for many commuters; therefore, these delays are more than just an inconvenience. Nice start. Delays impact work schedules, commute times, and overall public perceptions of the reliability of public transportation systems. This study aims to investigate and analyze TTC subway delay data from 2025 to determine (a) typically, I would have these letters capitalized, why are they in brackets? What type of delay occurs most frequently? (b) what type of delay causes the greatest impact, and (c) which type of delay is most likely preventable in the short term. Using Python for data analysis and visualization, delay codes were grouped into three general categories based on TTC delay data. The results of this analysis indicate that passenger-related delays are most common, while infrastructure-related delays have the greatest impact on overall delay times. A more in-depth analysis of passenger-related delays reveals that some of the largest contributing factors, such as disorderly patrons and unauthorized access to the track level, are more preventable than medical emergencies. This suggests that TTC should focus delay-reduction efforts not only on frequency, but also on impact and preventability.

Why transit reliability matters

Public transit reliability should be seen as an important issue because delays affect more than just travel schedules. An unreliable transit service may degrade the transit experience and create practical problems beyond arrival time. Studies on transit reliability show that unreliable service may lead to an irregular distribution of passengers across vehicles. As a result, some vehicles may be overcrowded while others are under-utilized, therefore affecting the quality of the transit experience (Zhang et al., 2020). In Toronto, these disruptions are especially significant because unplanned rail service interruptions can create substantial passenger delays and broader economic consequences for the city

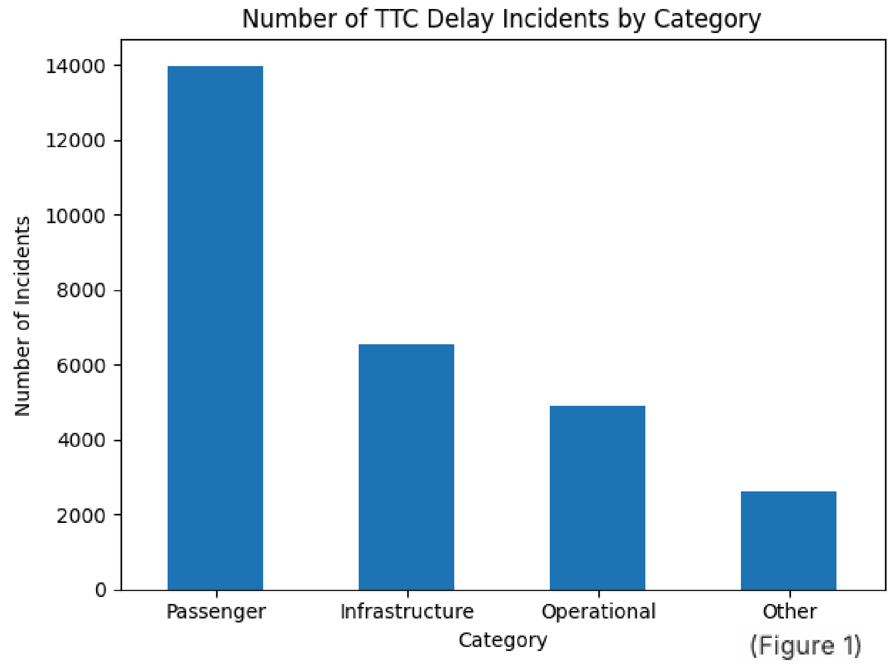
(Itani & Shalaby, 2021). In a city where many residents depend on the TTC for work, education, and daily activities, these disruptions can shape how passengers perceive the safety, quality of service, and reliability of public transportation. Therefore, delay data is not simply a log of transit disruptions; it also sheds light on how reliability issues impact the overall transit experience. In contrast, others may be better addressed through prevention/enforcement, rider education, and/or increased monitoring of access at the track level. Delayed data can also reveal broader issues of accountability, service quality, and the everyday experience of moving through the city.

Dataset and method

The primary data source for this study was the TTC Subway Delay Data, obtained from the City of Toronto Open Data portal; the accompanying code descriptions file was used to interpret subway delay codes (City of Toronto, n.d.). Using Python, the data were cleaned and merged to match subway delays with their corresponding code descriptions. The codes related to TTC subway delays were then grouped into four categories: passenger-related, operational, and infrastructure-related, and a residual Other category for codes that did not fit clearly within the main three groups. This allowed an evaluation not only of the frequency of subway delays but also of their overall impact on total delay time. After the initial analysis, passenger-related causes were further evaluated to identify those with the greatest impact and the highest short-term preventability. This was used to create a preventability matrix, in which causes such as disorderly patrons and unauthorized access to the track level were distinguished from less controllable causes, such as medical emergencies.

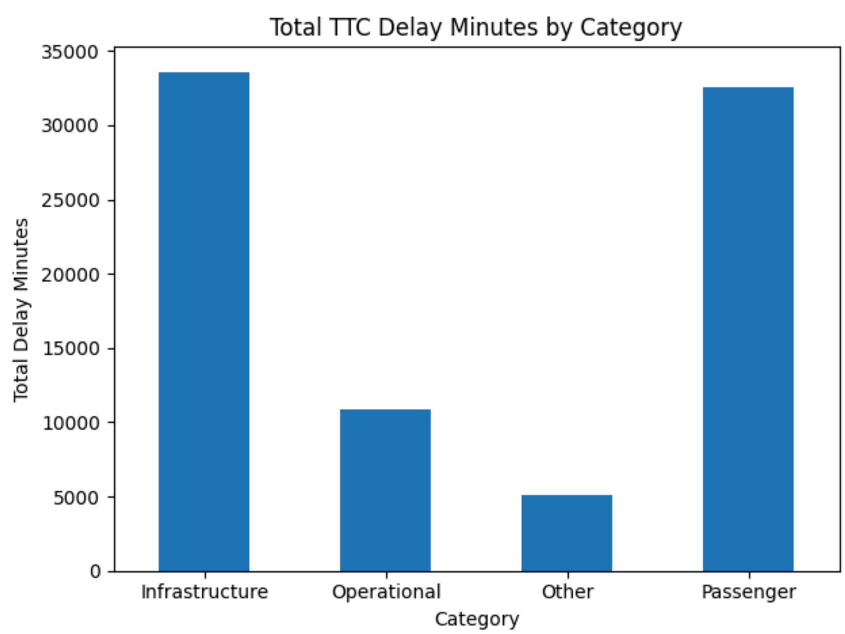
Main findings from the full dataset

The first stage of the analysis reveals a notable disparity between the frequency of TTC subway delays and the overall level of disruption. As depicted in **Figure 1,** passenger-caused delays are the most common category in the dataset, accounting for *13,976 delays*. Infrastructure-related delays are the second-most-common category, accounting for *6,555 delays.* Operational delays and the "Other" category comprise *4,907 and 2,609 delays*, respectively. Nonetheless, while passenger-caused delays are the most common, it is not possible to determine the overall impact of delays on the TTC subway solely from their frequency. **Figure 2** shows that infrastructure-related delays account for the greatest total delay time, totalling *33,585 minutes*. These delays are only slightly higher than those caused by passengers, reaching *32,548 minutes.*



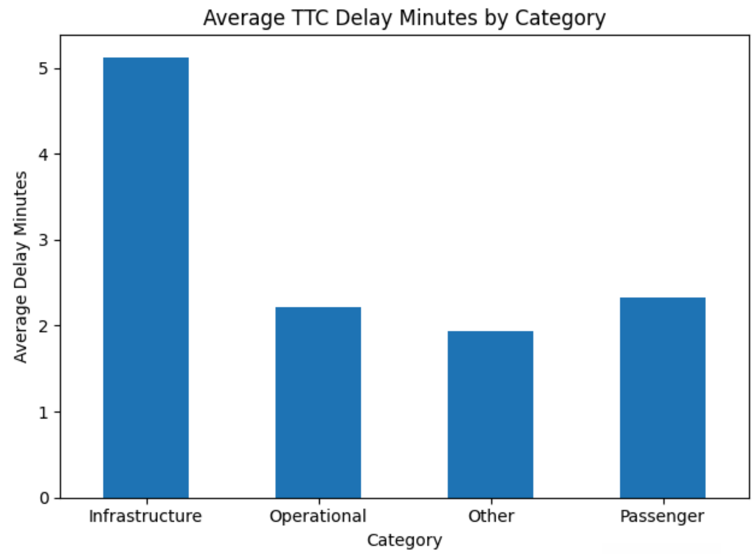
(Figure 1)

(Figure 2)



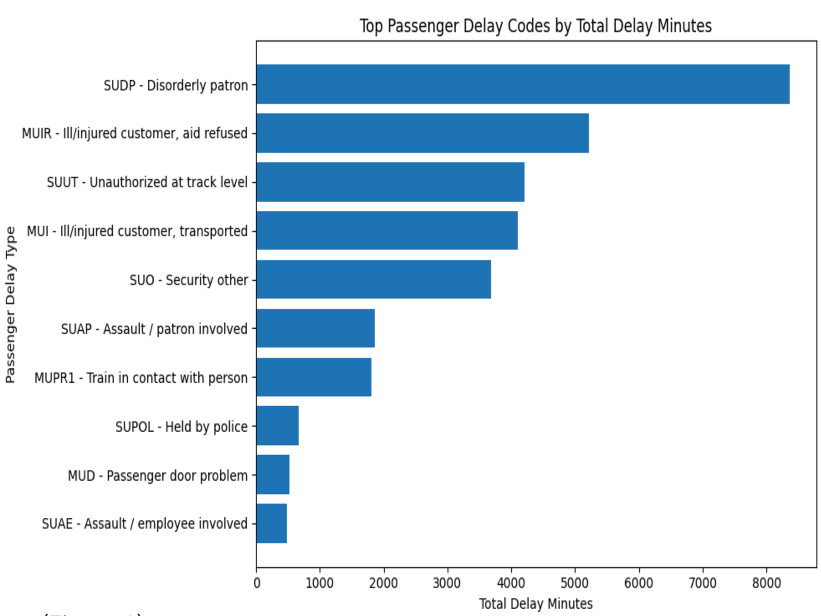
Furthermore, these delays are significantly higher than those caused by operational issues (*10,902 minutes)* and by "Other" *(5,057 minutes)*.

This reveals that although passenger delays are the most common, infrastructure delays have a greater impact on overall disruption in the TTC subway. **Figure 3** shows that infrastructure-related delays have the highest mean delay per incident, at *5.12 minutes*. This is significantly higher than delays caused by passengers (2.33 minutes), operational issues *(2.22 minutes)*, and "Other" *(1.94 minutes)*. Thus, it is evident that the overall impact of TTC subway delays cannot be understood solely in terms of their frequency.



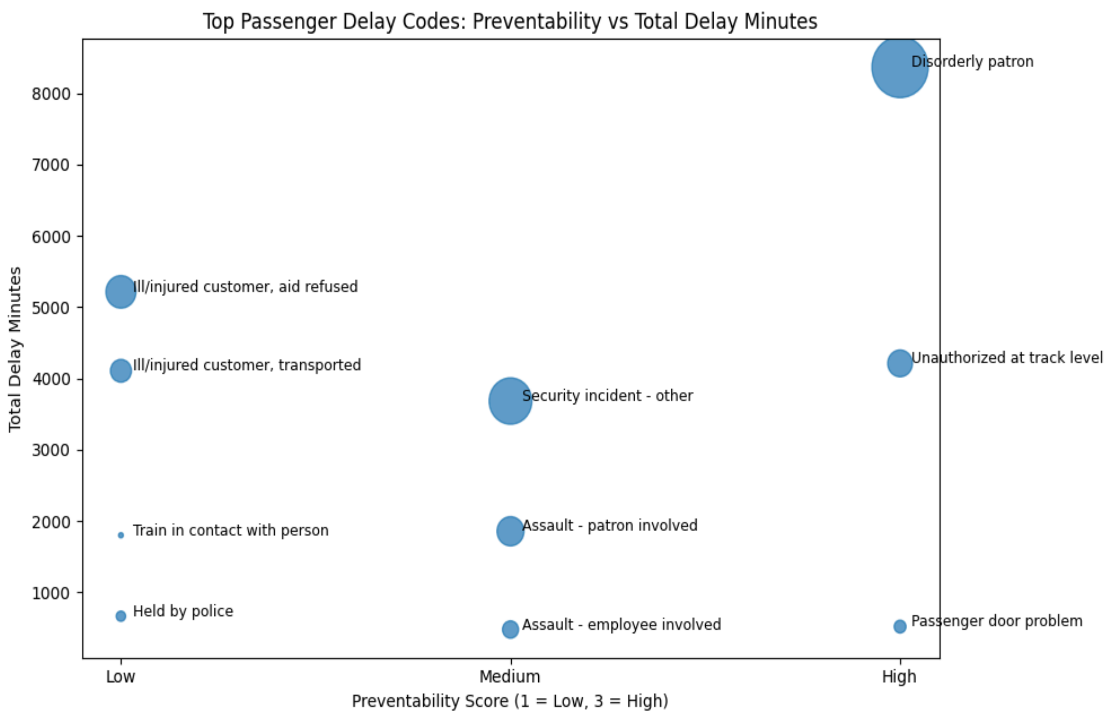
(Figure 3)

Passenger deep dive



(Figure 4)

A more detailed analysis of *passenger-related delays* reveals that this is not a single issue but a set of causes with varying levels of impact and preventability. As indicated in **Figure 4,** the passenger-related delay code that contributes most to total delay minutes is *disorderly patron*, followed by *ill/injured customer, aid refused, unauthorized access to track level, ill/injured customer, transported, and security incident* – other. It is evident from this figure that the most impactful passenger-related delays stem from a variety of causes, *including rider and medical-related incidents.* Alternatively, **Figure 5** indicates that the primary passenger-related causes identified in this dataset differ in their preventability. Although medical-related causes, such as ill or injured customers, contribute significantly to total delay minutes, they are less preventable than causes such as *disorderly patrons and unauthorized access to the track level*, which are more preventable *y*et still contribute substantially to delay minutes. This is an important finding, as reports from both the TTC and external sources suggest that rider-related causes, such as disorderly patrons and track-level access incidents, have become frequent and problematic sources of service disruption (Ouellet, 2017; McGillivray, 2025). This is especially significant given the broader effects of delays on the public, as TTC-related service disruptions have been described as affecting livelihoods, causing anxiety, and eroding trust in the transit system (McGillivray, 2025). Overall, this dataset suggests that causes such as disorderly patrons and unauthorized access to the track level may



(Figure 5)

be more suitable targets for short-term intervention than medical-related causes, even though both contribute significantly to passenger-related delay minutes.

References

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